VISHWATEJ KHOT

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PROFESSIONAL SUMMARY

AI Engineer and LLM specialist with 2+ years of industry experience designing and deploying production-grade RAG pipelines, intelligent chatbot systems, and multi-agent LLM workflows across UK-based companies. Skilled in vector databases (FAISS, ChromaDB, Pinecone), LangChain, HuggingFace Transformers, FastAPI, and AWS. Published researcher (ICETC 2025) currently pursuing an MSc in Data Science at the University of Bristol.

TECHNICAL SKILLS

RAG & Information Retrieval: RAG Pipelines, FAISS, ChromaDB, Pinecone, Sentence Embeddings, TF-IDF, Named Entity Recognition, Weaviate, Model Context Protocol

LLM Frameworks: LangChain, LlamaIndex, LangGraph, LangSmith, HuggingFace Transformers, Prompt Engineering, Model Fine-tuning

Languages & APIs: Python, FastAPI, REST APIs, SQL, Git, Docker, CI/CD

Cloud & MLOps: AWS (EC2, S3, CloudWatch), Model Deployment, Experiment Tracking, Model Monitoring, n8n, Power Automate

AI Domains: Generative AI, Multi-Agent Systems, Chatbot Development, Explainable AI, Computer Vision, Agentic AI

EMPLOYMENT HISTORY

AI Engineer | *Lume Studios Consulting, United Kingdom* Jan 2026 – Present

- Architected and deployed a production-grade, multi-client RAG system ingesting Google Drive documents (PDF, DOCX, XLSX, PPTX, CSV) via a 3-phase pipeline: parse, enrich with Claude Haiku context, and embed with OpenAI text-embedding-3-small using hybrid pgvector + tsvector search in Supabase, RRF fusion, and Cohere re-ranking; exposed as an MCP server so clients query their knowledge base from Claude Desktop, deployed on Hostinger VPS via Docker.
- Built a retrieval evaluation framework to benchmark chunk quality, reranking precision, and answer groundedness across client document sets, enabling systematic improvement of retrieval pipelines before client onboarding.
- Automated technical datasheet management by designing a Generative AI pipeline integrating LLMs to migrate from Excel-based systems to structured, queryable databases, reducing manual data retrieval time by 60%.

AI Engineer | *Abhinaya Ltd, United Kingdom* Nov 2024 – Aug 2025

- Architected and shipped DocAssist, a production Generative AI healthcare chatbot integrating LLMs and RAG for automated medical documentation and diagnostic decision support, reducing documentation time by 50%.
- Designed and implemented LangChain-based summarisation engines and structured knowledge modules using vector databases, enabling intelligent automation across dealflow and portfolio analysis systems.
- Built and maintained REST API endpoints using FastAPI to serve LLM-powered responses to downstream systems, with full end-to-end testing and deployment pipelines.

Generative AI Intern | *Intelimek Systems Pvt Ltd, Pune* Apr 2024 – Mar 2025

- Engineered a production-grade RAG system using LangChain and LLaMA to enable fast, accurate retrieval from large PDF corpora, reducing manual processing effort by 40%.
- Built NLP pipelines for document classification, summarisation, tagging, and structured data extraction, converting unstructured PDFs into searchable knowledge bases at scale.
- Integrated automated document analysis into existing enterprise workflows, improving consistency and throughput across cross-functional teams.

AI Intern | *Abhinaya Ltd, United Kingdom* May 2024 – Nov 2024

- Developed NeRF and Gaussian Splatting pipelines for 3D scene reconstruction from 4D datasets, improving volumetric rendering performance by 30%.
- Contributed to internal tooling and workflow documentation, building structured systems for model experimentation and iteration.

KEY PROJECTS

MedAssist AI — NHS Aligned Clinical Triage & RAG Knowledge System

- Architected a production-grade clinical decision support system using a hybrid RAG pipeline (FAISS + BM25 fused via Reciprocal Rank Fusion) with live confidence scoring, surfacing low-evidence retrievals below 40% to ensure clinicians know when the system is uncertain.
- Engineered a contradiction-aware ReAct agent loop that detects conflicting NICE guideline sources, triggers a refined re-search, and synthesises a reconciled answer rather than halting — ensuring reliable responses under real-world clinical ambiguity.
- Shipped full production reliability stack: XGBoost triage classifier on patient vitals (High/Medium/Low urgency), token bucket rate limiting with exponential backoff, 24-hour DiskCache eliminating redundant LLM calls, incremental MD5 dedup indexing, and end-to-end structured audit logging.

AegisAI — Self-Healing Enterprise RAG Chatbot

- Built a production-grade, self-healing RAG system for enterprise document Q&A using vector retrieval with confidence scoring, safety enforcement, and non-blocking recovery workflows, reducing hallucination failures by 40%.
- Delivered Flask API, real-time monitoring dashboards, and full audit logs; system ingests PDFs and unstructured documents and returns context-aware answers via LLM integration.

FinSight — Agentic Financial Research & Due-Diligence System

- Architected a 4-agent LangChain pipeline (Planner, Retriever, Contradiction, Memo) performing end-to-end financial due diligence across Lloyds annual reports, earnings transcripts, and filings, with every claim traced to its exact source file and page via provenance tracking and cross-document entity resolution.
- Built a production-grade RAG stack with FAISS vector search, incremental indexing with MD5 deduplication, token bucket rate limiting, 24-hour DiskCache, and MCP integration enabling external tools and data sources to plug into the agent pipeline via a standardised interface.
- Delivered a Contradiction Agent that cross-checks findings between sources and flags inconsistencies, a Memo Agent that writes a structured Buy/Hold/Sell investment memo, and a Streamlit UI where users upload PDFs and watch them sync into the live FAISS index in real time.

Multiclass Malware Detection (Modified GNN + Explainable AI)

- Built scalable feature pipelines transforming raw BYTE files into structured image representations for ML training, with Explainable AI reporting integrated with LLaMA-3.1 assisted analysis, reducing analyst investigation time by 45%.
- Research published at ICETC 2025 — DOI: 10.17492/jpi.pcet.022501

EDUCATION

MSc Data Science | *University of Bristol* — Sep 2025 – Present

B.E. Computer Engineering (CGPA: 8.69) | *SPPU University, Pune* — May 2021 – May 2025

Diploma in Data Science | *IIT Madras (Remote)* — Jan 2021 – Sep 2024

ACHIEVEMENTS & CERTIFICATIONS

- Research published at ICETC 2025 — DOI: 10.17492/jpi.pcet.022501
- IIT Madras Machine Learning Competition — 5th place
- Smart India Hackathon — 8th place (college level)
- Certifications: Machine Learning (Microsoft), Generative AI (Microsoft), Data Analysis (Google), Cloud Computing (AWS)